

BlazeDemo Flight Booking Automation Framework

Prepared By: Pranshu Mishra

Framework: Selenium WebDriver + Java + TestNG + Maven + POM

Project Type: Web Automation Testing

Repository: https://github.com/PRANSHU9999/Capstone_Project

Project Overview

The BlazeDemo Flight Booking Automation Framework is designed to automate the end-to-end flight booking workflow of the BlazeDemo web application.

The framework follows the Page Object Model (POM) architecture and incorporates TestNG, Maven, Extent Reports, Data-Driven Testing, Jenkins CI/CD integration, Screenshot Capture Mechanism, and Cross-Browser Testing capabilities.

The framework aims to reduce manual testing effort, improve test coverage, ensure repeatability, and provide detailed execution reporting.

Objective of the Project:

- Automate flight booking workflow
- Validate search and booking functionality
- Implement Data-Driven Testing
- Capture screenshots automatically
- Generate detailed reports
- Execute tests through Jenkins
- Demonstrate Cross-Browser Testing
- Identify application defects
- Follow industry-standard automation practice

Technologies Used

- Java
- Selenium WebDriver
- TestNG
- Maven
- Apache POI
- Extent Reports
- Jenkins
- Git
- GitHub
- Eclipse IDE
- Firefox
- Chrome
- Edge

Framework Architecture

Base Package

Contains:

- BaseTest.java

Responsible for:

- Browser Initialization
 - Browser Closure
 - Test Setup
-

Pages Package

Contains:

- HomePage.java
- FlightSelectionPage.java
- PurchasePage.java
- ConfirmationPage.java

Purpose:

Implementation of Page Object Model.

Utilities Package

Contains:

- ConfigReader.java
 - DriverFactory.java
 - ExcelUtils.java
 - ExtentManager.java
 - ScreenshotUtil.java
 - WaitUtils.java
-

Listeners Package

Contains:

- TestListener.java

Purpose:

- Extent Report Integration
 - Screenshot Capture on Failure
-

DataProviders Package

Contains:

- BookingDataProvider.java

Purpose:

Data-Driven Testing through Excel.

Automated Test Cases

LaunchTest

Objective:

Verify BlazeDemo launches successfully.

Expected Result:

Home Page displayed successfully.

SearchFlightTest

Objective:

Verify user can search flights.

Expected Result:

Flight results page displayed.

EndToEndBookingTest

Objective:

Verify complete booking workflow.

Flow:

Launch Application

↓

Search Flight

↓

Select Flight

↓

Enter Passenger Details

↓

Purchase Ticket

↓

Verify Confirmation

Expected Result:

Booking confirmation generated successfully.

7. Data-Driven Testing

Excel file is used to provide multiple booking datasets.

Fields:

- Source City
- Destination City
- Passenger Name
- Address
- City
- State
- Zip Code
- Card Number
- Card Type
- Name on the card

Benefits:

- Increased coverage
 - Reusable test scripts
 - Faster execution
-

8. Cross-Browser Testing

Framework supports:

Chrome

Primary execution browser.

Firefox

Cross-browser validation.

Edge

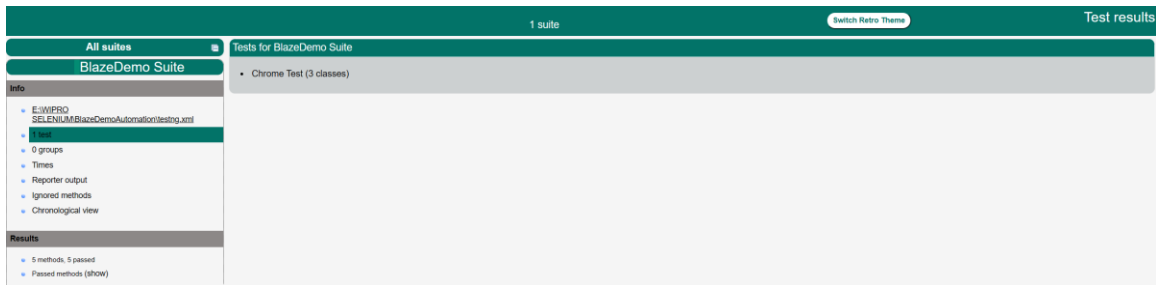
Cross-browser validation.

9. Reporting Mechanism

TestNG Reports

Location:

test-output/index.html



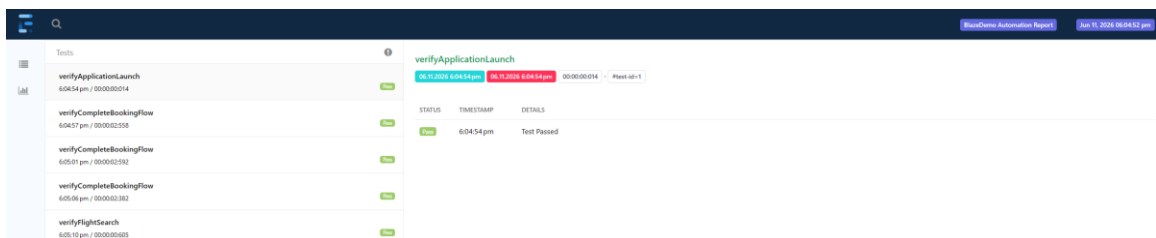
Provides:

- Passed Tests
- Failed Tests
- Skipped Tests
- Execution Summary

Extent Reports

Location:

reports/ExtentReport.html



Provides:

- Interactive Dashboard
- Pass/Fail Status
- Detailed Logs
- Screenshots
- Execution Timeline

Defect Suite Reports

Location:

test-output/BlazeDemo Defect Suite/

Defect Tests

Tests passed/Failed/Skipped:	3/0/0
Started on:	Sat Jun 06 21:02:59 IST 2026
Total time:	14 seconds (14732 ms)
Included groups:	
Excluded groups:	

(Hover the method name to see the test class name)

PASSED TESTS			
Test method	Attribute(s)	Exception	Time (seconds)
verifyBookingWithEmptyCardNumber Test class: defecttests.EmptyCardNumberTest		2	defecttests.EmptyCardNumberTest@2acadd4
verifyBookingWithExpiredCard Test class: defecttests.ExpiredCardTest		2	defecttests.ExpiredCardTest@24ad80c
verifyBookingWithEmptyPassengerDetails Test class: defecttests.EmptyPassengerDetailsTest		2	defecttests.EmptyPassengerDetailsTest@1165b38

Provides:

- Defect Validation Results
- Negative Test Case Execution
- Failed Validation Tracking

Cross-Browser Execution Reports

Location:

test-output/Cross Browser Suite/

Chrome Test

Tests passed/Failed/Skipped: 1/0/0	
Started on:	Thu Jun 11 18:04:28 IST 2026
Total time:	3 seconds (3313 ms)
Included groups:	
Excluded groups:	

(Hover the method name to see the test class name)

PASSED TESTS				
Test method	Attribute(s)	Exception	Time (seconds)	Instance
verifyApplicationLaunchInDifferentBrowsers		0	testcases.CrossBrowserLaunchTest	@2d1ef81a

Edge Test

Tests passed/Failed/Skipped: 1/0/0	
Started on:	Thu Jun 11 18:04:40 IST 2026
Total time:	2 seconds (2331 ms)
Included groups:	
Excluded groups:	

(Hover the method name to see the test class name)

PASSED TESTS			
Test method	Attribute(s)	Exception	Time (seconds)
verifyApplicationLaunchInDifferentBrowsers		0	testcases.CrossBrowserLaunchTest@791d18b6
Test class: testcases.CrossBrowserLaunchTest			

Firefox Test

Tests passed/Failed/Skipped: 1/0/0	
Started on:	Thu Jun 11 18:04:31 IST 2026
Total time:	9 seconds (9123 ms)
Included groups:	
Excluded groups:	

(Hover the method name to see the test class name)

PASSED TESTS			
Test method	Attribute(s)	Exception	Time (seconds)
verifyApplicationLaunchInDifferentBrowsers		0	testcases.CrossBrowserLaunchTest@4bbf6d0e
Test class: testcases.CrossBrowserLaunchTest			

Provides:

- Chrome Execution Results
- Firefox Execution Results
- Edge Execution Results

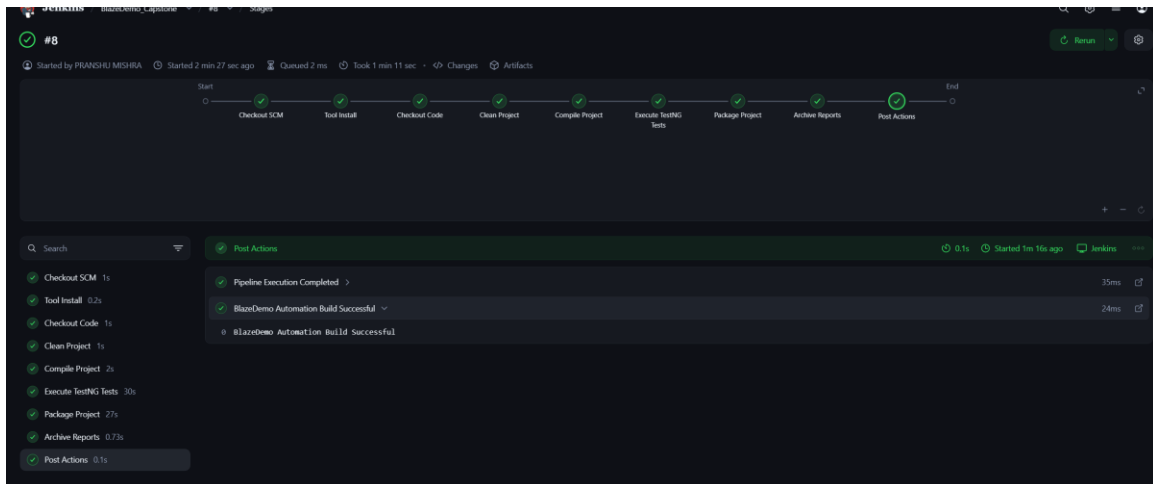
10. Screenshot Capture

Screenshots are automatically captured:

- On Test Failure
- During Important Workflow Steps

Benefits:

- Faster debugging
 - Defect evidence
 - Better reporting
-



Pipeline Stages:

1. Checkout Code
2. Clean Project
3. Compile Project
4. Execute TestNG Tests
5. Package Project
6. Archive Reports

Benefits:

- Automated Execution
 - Continuous Testing
 - Build Monitoring
-

12. Jira Epics and User Stories

12.1 Epic – User Flight Booking Journey (CB-1)

Capstone_BlazeDemo

Search timeline Epic Type Status category

Work

Sprints

CB-1 User Flight Booking Journey

- CB-2 Search Flights Between s...
- CB-3 View Flight Search Results
- CB-4 Purchase Selected Flight
- CB-5 View Booking Confirmation
- CB-23 Passenger Information ...
- CB-7 Website Navigation
- CB-8 Validate Home Page Nav...
- CB-9 Validate Destination Links
- CB-10 Form Validation
- CB-11 Mandatory Field Val...
- CB-12 Credit Card Validation
- CB-19 Name Field Accepts Onl...
- CB-20 Booking Companies W...
- CB-21 Expiration Month Field ...
- CB-22 Credit Card Number L...
- CB-24 Booking can be comple...
- CB-13 QA Automation & Test Management
- CB-14 Automation Framework ...
- CB-16 Test Automation ...
- CB-18 Reporting & CI/CD
- CB-17 Test Cases & Run...

CB-1 User Flight Booking Journey

Description

Objective

Enable a user to search for flights, select a flight, enter passenger details, complete a booking, and view the confirmation successfully.

Context

This issue describes the end-to-end booking flow that a user should be able to complete.

Scope

Included in this issue:

- Flight search
- Flight selection
- Passenger details entry
- Booking completion
- Confirmation display

Not included:

- N/A

Success criteria

- A user can search flights.
- A user can select a flight.
- A user can enter passenger details.
- A user can complete the booking.
- A user can view the confirmation successfully.

Risks and dependencies

- N/A

Other information

- Issue Name: Create a Jira Project Structure

User Story – Search Flights Between Source and Destination (CB-2)

Capstone_BlazeDemo

Search timeline Epic Type Status category

Work

Sprints

CB-1 User Flight Booking Journey

- CB-2 Search Flights Between s...
- CB-3 View Flight Search Results
- CB-4 Purchase Selected Flight
- CB-5 View Booking Confirmation
- CB-23 Passenger Information ...
- CB-7 Website Navigation
- CB-8 Validate Home Page Nav...
- CB-9 Validate Destination Links
- CB-10 Form Validation
- CB-11 Mandatory Field Val...
- CB-12 Credit Card Validation
- CB-19 Name Field Accepts Onl...

CB-2 Search flights between source and destination cities

Description

User story

As a user, when I want to search for available flights, I want to choose a source city and a destination city so that I can find a suitable flight.

Context

The issue is about searching for flights between a source city and a destination city.

Acceptance criteria

- Source city dropdown is visible
- Destination city dropdown is visible
- User can select source city
- User can select destination city
- Find Flights button is clickable
- Flight results page is displayed

Other information

N/A

User Story – View Flight Search Results (CB-3)

The screenshot shows the Jira interface for the 'Capstone_BlazeDemo' project. The 'Timeline' view is selected, showing a sprint plan for the current sprint. The user story 'CB-3 View Flight Search Results' is highlighted in blue. The right-hand panel displays the details for this user story, including its description, context, acceptance criteria, and other information.

User Story Details:

- Title:** View Flight Search Results
- Description:** User story
- Context:** As a user, when I search for flights, I want to view available flights so that I can choose a flight.
- Acceptance criteria:**
 - Flight table is displayed
 - Flight information is visible
 - Flight price is displayed
 - Choose This Flight button is available
 - User can select any flight
- Other information:** N/A

User Story – Purchase Selected Flight (CB-4)

The screenshot shows the Jira interface for the 'Capstone_BlazeDemo' project. The 'Timeline' view is selected, showing a sprint plan for the current sprint. The user story 'CB-4 Purchase Selected Flight' is highlighted in blue. The right-hand panel displays the details for this user story, including its description, context, acceptance criteria, and other information.

User Story Details:

- Title:** Purchase Selected Flight
- Description:** User story
- Context:** As a user, when I want to enter my booking details, I want to purchase a flight so that my booking is successfully completed.
- Acceptance criteria:**
 - Home field available
 - Address field available
 - City field available
 - State field available
 - Zip code field available
 - Credit card fields available
 - Purchase button available
 - Booking successfully completed
- Other information:** N/A

User Story – View Booking Confirmation (CB-5)

The screenshot shows the Jira interface for the 'Capstone_BlazeDemo' project. The 'Timeline' view is selected, showing a sprint plan for the current sprint. The user story 'CB-5 View Booking Confirmation' is highlighted in blue. The right-hand panel displays the details for this user story, including its description, context, acceptance criteria, and other information.

User Story Details:

- Title:** View Booking Confirmation
- Description:** User story
- Context:** As a user, when I have completed a purchase, I want to see booking confirmation details so that I know my booking was successful.
- Acceptance criteria:**
 - Confirmation page displayed
 - Booking ID generated
 - Booking status displayed
 - Amount displayed
 - User receives successful booking message
- Other information:** N/A

12.2 Epic – Website Navigation (CB-7)

The screenshot displays the Jira interface for the 'Capstone_BlazeDemo' project. The 'Epic' view is selected, showing a list of tasks under the 'Website Navigation' epic. The tasks are: CB-1 User Flight Booking Journey, CB-2 Search Rights between s..., CB-3 View Flight Search Results, CB-4 Purchase Selected Flight, CB-5 View Booking Confirmation, CB-6 Passenger Information..., CB-7 Website Navigation, CB-8 Validate Home Page Navig..., CB-9 Validate Destination Links, CB-10 Form Validation, CB-11 Mandatory Field Val..., and CB-12 Credit Card Validation. The 'Website Navigation' epic is highlighted in green. The right sidebar shows the details for the selected epic, including its description, objective, context, scope, success criteria, risks, and dependencies.

Website Navigation

Description
Objective
Improve the website navigation.
Context
The issue is titled "Website Navigation".
Scope
• In scope: website navigation.
• Out of scope: N/A.
Success criteria
• Website navigation has been improved.
Risks and dependencies
• Risks: N/A.
• Dependencies: N/A.
Other information
Website Navigation

User Story – Validate Home Page Navigation (CB-8)

The screenshot displays the Jira interface for the 'Capstone_BlazeDemo' project. The 'User Story' view is selected, showing the details for the 'Validate Home Page Navigation' user story. The story is highlighted in green. The right sidebar shows the details for the selected user story, including its description, user story, context, acceptance criteria, risks, and dependencies.

Validate Home Page Navigation

Description
User story
As a user, when I access the home page, I want to navigate through the available links so that I can reach all pages without issues.
Context
This issue is about validating home page navigation.
Acceptance criteria
• Home page loads successfully
• Navigation links visible
• No broken links
• User can access all pages
Other information
N/A

User Story – Validate Destination Links (CB-9)

The screenshot displays the Jira interface for the 'Capstone_BlazeDemo' project. The 'User Story' view is selected, showing the details for the 'Validate Destination Links' user story. The story is highlighted in green. The right sidebar shows the details for the selected user story, including its description, user story, context, acceptance criteria, risks, and dependencies.

Validate Destination Links

Description
User story
As a user, when I open a destination link, I want the destination page to load successfully so that I can confirm the link works.
Context
The issue is to validate destination links.
Acceptance criteria
• Destination pages open successfully
• Correct page title displayed
• No HTTP errors
Other information
N/A

12.3 Epic – Form Validation (CB-10)

Form Validation

Description
Objective
Validate that data entered into the form is correct.

Context
The issue is about form validation.

Scope
• Check whether data is entered correctly in the form.

Success criteria
• The form validates entered data correctly.

Risks and dependencies
N/A

Other information
Form validation to check if data is entered correctly in the form.

Child work items

Work	PH...	As...	Status
CB-10 Form Validation	M	M	In Progress
CB-11 Mandatory Field Validation	M	M	Not Started
CB-12 Credit Card Validation	M	M	Not Started
CB-13 Name Field Accepts Only Special Characters	M	M	Not Started
CB-14 Booking Completes Without Entering City and State	M	M	Not Started
CB-15 Expiration Month Field Accepts Invalid Values	M	M	Not Started
CB-16 Credit Card Number Length Not Validated	M	M	Not Started
CB-17 Booking can be completed without entering	M	M	Not Started

User Story – Mandatory Field Validation (CB-11)

Mandatory Field Validation

Description
User story
As a user, when I leave mandatory fields empty, I want to see validation so that I cannot proceed with incomplete data.

Context
This issue is about mandatory field validation.

Acceptance criteria
• Empty mandatory fields show validation.
• User cannot proceed with incomplete data.

Other information
N/A

Subtasks
None entered

User Story – Credit Card Validation (CB-12)

Credit Card Validation

Description
User story
As a user, when I enter a credit card number, I want invalid card numbers to be rejected so that I know the card details are not valid.

Context
This issue is about credit card validation.

Acceptance criteria
• Invalid card numbers rejected.
• Empty card field validation displayed.
• Error message shown.

Other information
N/A

12.4 Epic – QA Automation & Test Management (CB-13)

QA Automation & Test Management

Description
Objective
Design, automate, execute, and report test cases for the BlazeDemo application.

Context
This work is focused on QA automation and test management for the BlazeDemo application.

Scope

- Design test cases for the BlazeDemo application.
- Automate the test cases.
- Execute the test cases.
- Report on the test cases.

Success criteria

- Test cases are designed for the BlazeDemo application.
- Test cases are automated.
- Test cases are executed.
- Test results are reported.

Risks and dependencies

- Dependency on the BlazeDemo application being available for testing.

Other information
N/A

Child work items
50% Done

- CB-14 Automation Framework Setup
- CB-15 Test Automation Development
- CB-16 Reporting & CI/CD

User Story – Automation Framework Setup (CB-14)

Automation Framework Setup

Description
User story
As a tester, when setting up the automation framework, I want to create a Maven project, configure Selenium and TestNG, implement the Page Object Model, create a base test and utilities, and configure logging so that the framework is ready for automated testing.

Context
The issue is about setting up an automation framework.

Acceptance criteria

- Create a Maven project.
- Configure Selenium and TestNG.
- Implement the Page Object Model.
- Create a base test and utilities.
- Configure logging.

Other information

- Issue name: Automation Framework Setup

Subtasks
Add subtask

Linked work items
Add linked work item

Confidence content
100% (Template)

User Story – Test Automation Development (CB-15)

Test Automation Development

Description
User story
As a user, when I search for flights, I want the flight search to be automated so that I can complete the search process efficiently.

Context
This issue is about developing test automation for the flight booking flow.

Acceptance criteria

- Automate flight search.
- Automate flight selection.
- Automate purchase flow.
- Automate confirmation validation.

Other information

Tasks

- Automate flight search
- Automate flight selection
- Automate purchase flow
- Automate confirmation validation

Subtasks
Add subtask

User Story – Reporting & CI/CD Integration (CB-16)

The screenshot displays the Jira interface for a project named "Capstone_BlazeDemo". The left sidebar shows the project structure with "QA Testing" selected. The main panel shows a list of tasks under the "CB-16" epic, including "CB-1 User Flight Booking Journey", "CB-2 Search flights between...", "CB-3 View flight search results...", "CB-4 Purchase selected flight...", "CB-5 View booking confirmation...", "CB-6 Passenger information...", "CB-7 Website navigation...", "CB-8 Validate home page nav...", "CB-9 Validate destination links...", "CB-10 Form validation...", "CB-11 Mandatory field val...", "CB-12 Credit card validation...", "CB-13 Name field accepts O...", "CB-14 Booking completion W...", "CB-15 Expiration month field...", "CB-16 Credit card number L...", "CB-17 Booking can be comple...", "CB-18 QA Automation & Test Management...", "CB-19 Automation Framework...", "CB-20 Test Automation...", "CB-21 Reporting & CI/CD...", and "CB-22 Test Design & Execution...". The right panel shows the details for "CB-16: Reporting & CI/CD".

User story

As a user, when I run automated tests, I want to generate execution reports so that I can review the test results.

As a developer, when the repository and CI/CD process are set up, I want to configure GitHub and Jenkins so that reporting and test execution are integrated.

Acceptance criteria

- Event Reports are integrated.
- The GitHub repository is configured.
- A Jenkins pipeline is set up.
- Execution reports are generated.

Other information

Tasks:

- Integrate Event Reports
- Configure GitHub Repository
- Setup Jenkins Pipeline
- Generate Execution Reports

Subtasks

Add subtask

Linked work items

Add linked work item

User Story – Test Design & Execution (CB-17)

The screenshot displays the Jira interface for a project named "Capstone_BlazeDemo". The left sidebar shows the project structure with "QA Testing" selected. The main panel shows a list of tasks under the "CB-17" epic, including "CB-1 User Flight Booking Journey", "CB-2 Search flights between...", "CB-3 View flight search results...", "CB-4 Purchase selected flight...", "CB-5 View booking confirmation...", "CB-6 Passenger information...", "CB-7 Website navigation...", "CB-8 Validate home page nav...", "CB-9 Validate destination links...", "CB-10 Form validation...", "CB-11 Mandatory field val...", "CB-12 Credit card validation...", "CB-13 Name field accepts O...", "CB-14 Booking completion W...", "CB-15 Expiration month field...", "CB-16 Credit card number L...", "CB-17 Booking can be comple...", "CB-18 QA Automation & Test Management...", "CB-19 Automation Framework...", "CB-20 Test Automation...", "CB-21 Reporting & CI/CD...", and "CB-22 Test Design & Execution...". The right panel shows the details for "CB-17: Test Design & Execution".

User story

As a tester, when preparing for testing, I want to create smoke test cases and regression test cases so that I can execute the test suite, log defects, and reset fixed defects.

Acceptance criteria

- Create smoke test cases.
- Create regression test cases.
- Log defects.
- Reset fixed defects.

Other information

N/A

Subtasks

Add subtask

Linked work items

Add linked work item

Confidence content

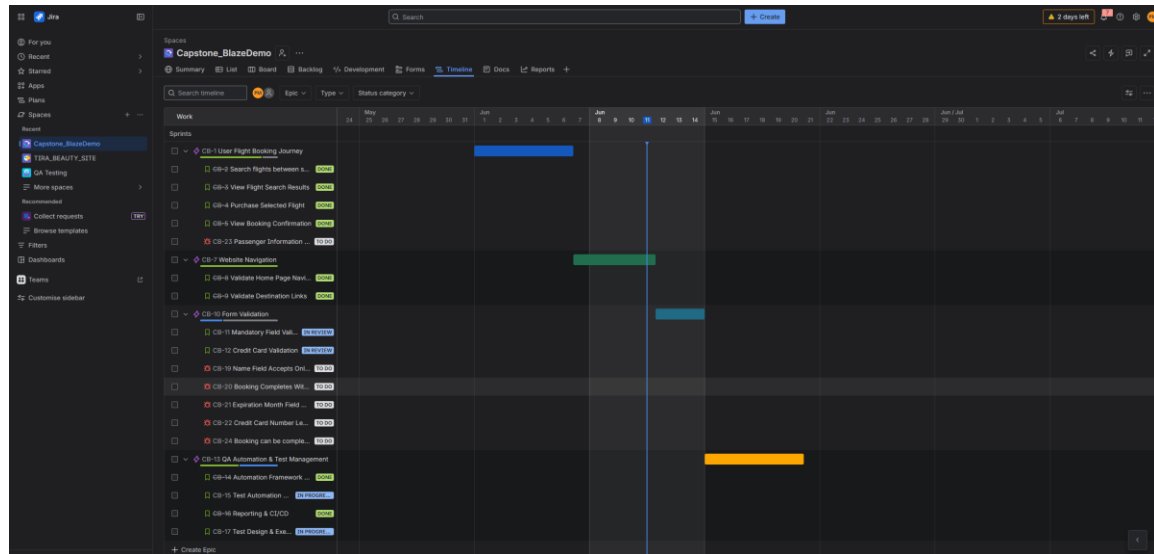
Product requirements

Details

Assignee: Unassigned

Assign to me

12.5 Jira Timeline Overview



13. Defects Report

Defect 1

Name field accepts special characters.

The screenshot shows the Jira work item details for defect CB-19, titled 'Name Field Accepts Only Special Characters'. The interface includes a sidebar with navigation options like 'For you', 'Recent', 'Starred', 'Apps', 'Plans', 'Spaces', and 'Teams'. The main area displays the details of the defect, including its description, impact, expected behavior, actual behavior, steps to reproduce, environment, and workarounds.

Name Field Accepts Only Special Characters

To Do ▾ ⚙️ ⚡ Improve Bug

Description

Impact

The Name field accepts only special characters. This allows a booking to proceed with invalid name input. Users are not shown a validation error.

Expected behaviour

The system should reject special characters in the Name field and display a validation message.

Actual behaviour

Booking proceeds successfully with special characters entered as name.

Steps to reproduce

- Open BlazeDemo.
- Search and select a flight.
- Navigate to Purchase Flight page.
- Enter "q!a!\$%&*~" in Name field.
- Complete remaining mandatory fields.
- Click Purchase Flight.

Environment

- Windows 11
- Chrome Latest

Workarounds

N/A

Defect 2

Booking completes without City and State.

The screenshot shows the Jira interface for the 'Capstone_BlazeDemo' project. The 'Timeline' view is active, displaying a backlog of items. The item 'CB-20 Booking Completes Without Entering City and State' is highlighted. The right-hand panel shows the details for this issue, including its description, impact, expected behavior, actual behavior, steps to reproduce, environment, and workarounds.

Jira work item

CB-20 / CB-20

To Do ⚙️ + Improve Bug

Description

Impact
High severity issue. Users can complete a booking without entering City and State.

Expected behaviour
A validation message should appear requiring City and State.

Actual behaviour
Booking completes successfully.

Steps to reproduce

1. Open BlazeDemo.
2. Search and select a flight.
3. Enter Name and Address.
4. Leave City and State blank.
5. Complete remaining fields.
6. Click Purchase Flight.

Environment
BlazeDemo

Workarounds
N/A

Other information
Severity: High

Defect 3

Invalid expiry month values accepted.

The screenshot shows the Jira interface for the 'Capstone_BlazeDemo' project. The 'Timeline' view is active, displaying a backlog of items. The item 'CB-21 Expiry Month Field Accepts Invalid Values' is highlighted. The right-hand panel shows the details for this issue, including its description, impact, expected behavior, actual behavior, steps to reproduce, environment, and workarounds.

Jira work item

CB-21 / CB-21

To Do ⚙️ + Improve Bug

Description

Impact
The Expiry Month field accepts invalid values. This can let users submit incorrect month information on the Purchase Flight page.

Expected behaviour
Only values between 01 and 12 should be accepted.

Actual behaviour
Invalid month values are accepted.

Steps to reproduce

- Open Purchase Flight page.
- Enter valid passenger details.
- Enter month values such as:
 - + 00
 - + 13
 - + 25
 - + 99
- Click Purchase Flight.

Environment
Purchase Flight page

Other information
• Severity: Medium

Defect 4

Credit card validation is weak.

The screenshot shows the Jira interface for a project named 'Capstone_BlazeDemo'. The 'Timeline' view is selected, showing a list of tasks. A task titled 'CB-22 Credit Card Number Length Not Validated' is highlighted. The right-hand panel displays the details of this task, including its description, impact, expected behavior, actual behavior, steps to reproduce, and other information.

Task Details:

- Title:** Credit Card Number Length Not Validated
- Description:** Users can complete a booking with invalid credit card numbers. This may allow incorrect payment details to pass through validation.
- Impact:** The system should validate the card length and reject invalid values.
- Expected behaviour:** Booking completes successfully.
- Actual behaviour:** Booking completes successfully.
- Steps to reproduce:**
 - Open Purchase Flight page.
 - Enter passenger details.
 - Enter:
 - 123
 - 123456
 - 12345678901234567890
 - Submit booking.
- Other information:**
 - Severity: High

Defect 5

Passenger information mismatch on confirmation page.

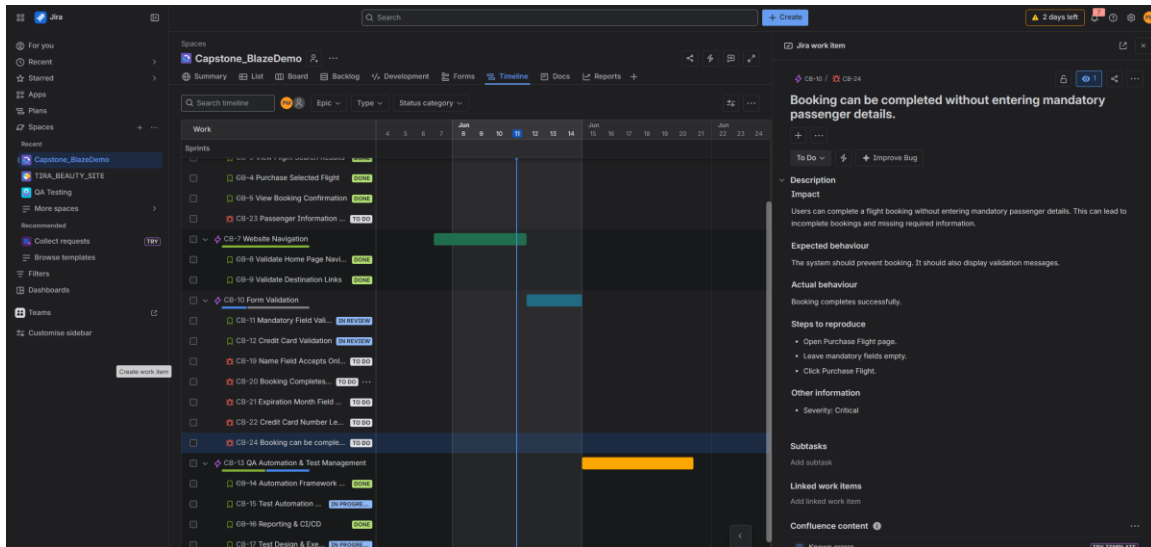
The screenshot shows the Jira interface for a project named 'Capstone_BlazeDemo'. The 'Timeline' view is selected, showing a list of tasks. A task titled 'CB-23 Passenger Information Mismatch on Confirmation Page' is highlighted. The right-hand panel displays the details of this task, including its description, impact, expected behavior, actual behavior, steps to reproduce, and other information.

Task Details:

- Title:** Passenger Information Mismatch on Confirmation Page
- Description:** Passenger information on the confirmation page does not match the submitted details. This can affect users who need to verify their booking information.
- Impact:** The confirmation page should display the exact information submitted by the user.
- Expected behaviour:** Passenger information differs from the submitted values.
- Actual behaviour:** Passenger information differs from the submitted values.
- Steps to reproduce:**
 - Search and select flight.
 - Enter passenger details.
 - Complete booking.
 - Compare entered data with confirmation page.
- Other information:**
 - Severity: Critical

Defect 6

Booking completes without mandatory passenger details.



14. Challenges Faced

Challenge 1

Edge browser execution instability in Jenkins.

Solution

Configured Jenkins to execute Chrome tests while maintaining separate Cross-Browser suite.

Challenge 2

Git merge conflicts during framework updates.

Solution

Resolved Jenkinsfile and pom.xml conflicts manually.

Challenge 3

Maintaining reusable framework architecture.

Solution

Implemented Page Object Model and Utility classes.

15. Project Achievements

- ✓ Complete Selenium Framework Developed
 - ✓ Page Object Model Implemented
 - ✓ Data-Driven Testing Implemented
 - ✓ Extent Reports Integrated
 - ✓ Screenshot Capture Implemented
 - ✓ Jenkins CI/CD Integrated
 - ✓ Cross-Browser Testing Implemented
 - ✓ GitHub Version Control Maintained
 - ✓ Defect Validation Suite Developed
-

16. Conclusion

The BlazeDemo Flight Booking Automation Framework successfully automates the end-to-end flight booking process using Selenium WebDriver, Java, TestNG, and Maven. The framework follows industry-standard automation practices including Page Object Model (POM), Data-Driven Testing, Extent Reporting, Jenkins CI/CD integration, Screenshot Capture, and Cross-Browser Testing.